

INSPIRE MANAK Awards 2026

1. Project Title & Basic Information

- **Project Title:** Smart Community Water Monitoring: An IoT-Based System for Detecting Pipeline Leakages and Illegal Water Theft
- **Program:** INSPIRE Awards - MANAK (Million Minds Augmenting National Aspirations and Knowledge) Project Proposal
- **Target Area:** Water Resource Management & Community Infrastructure Protection
- **School Details:** St. John Vianney School, Gwalior, Madhya Pradesh

2. Abstract & Introduction

Public water supply systems form the bedrock of urban and rural infrastructure, yet they face severe efficiency losses due to undetected pipeline damage, unmonitored usage, and unauthorized extractions. Traditional monitoring approaches rely on periodic manual inspections, which often identify leaks only after substantial water loss or visible ground subsidence has occurred. Furthermore, municipal authorities frequently struggle to track off-peak wastage, such as open residential taps or overflowing overhead tanks during non-peak hours, as well as illegal taps along distribution mains.

To address these challenges, this project introduces an integrated, IoT-based three-in-one monitoring solution engineered to deliver comprehensive oversight across localized distribution zones:

- **Underground Pipeline Leakage Detection:** Automatically spots when water disappears between the main inlet and delivery points due to broken or cracked public pipes, preventing road cave-ins and severe resource loss.
- **Water Theft (Illegal Tapping) Prevention:** Identifies unauthorized diversions and illegal hookups when the total water entering a zone does not match the sum of authorized household readings.
- **Night-Time Open Tap & Overflow Monitoring:** Tracks continuous, uninterrupted flow during off-peak night hours when domestic consumption should be zero, catching forgotten open taps, tank overflows, or running leaks and sending direct warnings to both the homeowner and local authorities.

By providing continuous automated oversight, this framework transforms reactive maintenance into proactive resource management for local water infrastructure.

3. Components & Materials Required

The physical prototype serves as a cost-effective proof of concept for local validation, whereas a full-scale municipal deployment would necessitate upgrading to robust, industrial-grade components designed for demanding outdoor environments.

Component Name	Prototype Use	Future Industrial Upgrade
Microcontroller / Controller	We use the ESP32 microcontroller as the main processing unit of our model. It reads flow counts from the sensors, performs the comparison calculations, and connects to local Wi-Fi to transmit alert messages.	Rugged, industrial-grade telemetry controller equipped with LoRaWAN or GSM cellular modules. Since street pipelines lack Wi-Fi, this enables long-range wireless data transmission over several kilometers directly to municipal servers.
Water Flow Sensors	We use small, plastic YF-S201 hall-effect water flow sensors. They contain a turbine that spins as water passes, allowing us to measure flow rates in our small-scale test pipes.	High-precision, non-intrusive industrial ultrasonic or magnetic flow meters. These are designed to withstand high pressure, prevent physical wear, and fit large-diameter municipal water mains.
Control Valves	We use a small motorized solenoid valve in the prototype to demonstrate automatic flow cutoff whenever a leak or mismatch is detected.	Heavy-duty motorized actuator valves. These industrial-grade valves have high torque to turn off large municipal pipelines safely and provide feedback to verify the valve's status.

Component Name	Prototype Use	Future Industrial Upgrade
Relay Module	We use a standard 5V relay board to allow the low-power microcontroller to switch and control the power going to the motorized valve.	Industrial solid-state relays (SSRs) or Programmable Logic Controllers (PLCs) mounted in protective panels for sparks-free, safe, and highly reliable switching.
Power Supply	We power our prototype using a standard USB cable and a 5V power adapter for indoor testing and programming.	Off-grid solar power system pairing solar panels with rechargeable LiFePO4 batteries, ensuring continuous, self-sustaining operation in remote outdoor locations.
Protective Enclosure	We house our prototype wiring and boards in a basic plastic project box to keep the connections organized.	IP67/IP68 weatherproof, dust-tight, and waterproof enclosures designed for harsh outdoor installations, equipped with a GPS module to broadcast exact coordinate locations.
Custom Mobile/PC Application	A simple application used to view live status updates and receive instant alerts.	Centralized municipal dashboard featuring advanced analytics, automated reporting, and historical trend analysis.
Firebase Cloud Database	Real-time cloud synchronization service to push alerts, stream data, and coordinate action immediately.	Secure, scalable enterprise-level cloud infrastructure designed for continuous real-time monitoring and robust data security.

Cost-Efficiency Budget Summary

The prototype relies on highly accessible, cost-effective off-the-shelf microelectronics. This shows that real-time zonal water monitoring is achievable with low initial infrastructure investments, presenting a practical and highly scalable framework for municipalities.

4. Working Principle and Method

The core idea behind our system is maintaining a simple "water balance" for a specific zone of the pipeline. We place a main flow sensor at the entry point where water enters a neighborhood zone and smaller flow sensors at every delivery point, such as individual houses or apartment blocks.

The system continuously compares the volume of water entering the zone against the total amount of water received by all connected delivery points:

- **Normal Operation:** If the water entering the zone matches the total water recorded at all houses, the system assumes everything is working correctly. Even if someone uses an electric pump to draw more water, the balance holds because the household sensors will register that extra usage.
- **Leak or Theft Detection:** If the water entering the zone is significantly higher than the total water reaching the households, the system immediately recognizes a mismatch. Because this missing water is not reaching any authorized destination, the system concludes that it must be leaking out through a broken pipe underground or being stolen through an illegal, unauthorized tap.
- **Automated Response:** As soon as this mismatch is detected, the controller triggers an instant alert. This notification is sent directly to a custom mobile app used by municipal or local authorities. With this immediate warning, the authorities can quickly locate the problem area, send a team to fix the leak, or shut off the water supply to that specific segment to prevent further loss until the repairs are completed.

Note: The architecture diagrams and flowcharts shown below are temporary conceptual previews. These will be replaced with complete, hand-sketched schematics currently being drawn on a digital tablet.

Step-by-Step Workflow

- **Step 1: Primary Water Inflow Measurement:** Water entering a target distribution zone passes through a primary flow meter installed at the main inlet, continuously recording the total volume supplied to the area.
- **Step 2: Consumption Tracking at Delivery Points:** Secondary flow meters located at individual household connections measure the exact volume of water delivered to authorized end users.
- **Step 3: Real-Time Firebase Cloud Syncing:** The central controller collects flow rate readings from both the primary and secondary sensors and transmits the data to the Firebase cloud database over a wireless network.
- **Step 4: Automated Mismatch Analysis:** An automated algorithm calculates the total water balance by subtracting the sum of consumer usages from the total inlet volume. A discrepancy exceeding standard tolerance limits indicates an underground leak or unauthorized tap.
- **Step 5: Instant Mobile Alert & Valve Control:** When a leak or theft condition is identified, the system immediately transmits a notification to the municipal mobile application and actuates line control valves to isolate the damaged pipe section.
- **Step 6: Night-Time Open Tap & Continuous Flow Monitoring:** During off-peak night hours when household consumption is normally zero, the system checks for continuous, uninterrupted water flow. If a household flow sensor detects running water for an unusually long duration, the system concludes that a tap has been left open or a tank is overflowing, and immediately triggers a dual response: sending a direct warning notification to the resident's mobile app or via SMS, while also alerting local authorities to prevent severe water wastage.
- **Step 7: Power Stability & Circuit Resilience:** To protect against local electrical noise and minor voltage dips, dedicated power regulation circuitry keeps the controller running smoothly and ensures the sensors give accurate, steady readings.

5. National Impact & Real-World Problem Solution Matrix

Indian municipal water distribution networks face systemic operational and financial hurdles that severely impede sustainable urban development. Deploying our

automated IoT monitoring framework directly addresses these structural challenges while supporting overarching national infrastructure goals.

S.No	India Level Problem	National Data / Proof	How Our Project Solves It
1	High Non-Revenue Water (NRW) Losses & Underground Pipeline Leakages	Major Indian cities suffer from devastating Non-Revenue Water (NRW) losses ranging between 30% and 50% due to unmonitored underground pipe leaks. Cities like Mumbai and Delhi lose hundreds of millions of liters of treated water daily before it ever reaches consumers, causing massive physical and economic resource depletion.	The system deploys dual YF-S201 Hall-effect water flow sensors (one main inlet sensor acting as the system's "sensory nerve" and secondary household sensors). The ESP32 microcontroller (acting as the system's "brain") continuously runs a real-time volumetric water-balance algorithm. If the water volume passing through the main inlet exceeds the sum of the volumes recorded at the authorized consumer points, a mismatch is immediately flagged, revealing underground leaks.
2	Unauthorized Water Diversions & Water Theft (Illegal Tapping)	Unauthorized connections and illegal water tapping along distribution mains severely drain municipal supplies, leading to acute water scarcity in planned neighborhoods. In metropolitan areas like Bengaluru, unbilled consumption and unauthorized extractions	The system continuously monitors the water balance of the zone. If an unauthorized diversion occurs, the total water volume entering the zone does not match the sum of authorized consumer meters. When this deficit exceeds the safety tolerance threshold, the

S.No	India Level Problem	National Data / Proof	How Our Project Solves It
		lead to massive supply imbalances and hinder equitable water distribution.	ESP32 acts immediately by sending a command to a 12V solenoid valve (the "hands" of the system) to shut off the supply to the affected pipe section in just 2-3 seconds, preventing further theft while sounding a local buzzer for immediate on-site alert.
3	Severe Financial & Revenue Losses for Municipal Corporations	Unaccounted and unbilled water results in staggering financial losses for urban local bodies. Municipal corporations collectively suffer massive annual financial drains, with some metropolitan authorities experiencing losses up to ₹4,500 crore (cr) annually due to non-revenue water and inefficient billing.	By providing 100% accurate, real-time digital consumption tracking, our system eliminates human error, manual inspection delays, and billing discrepancies. The ESP32 synchronizes all flow data instantly to the cloud via Firebase. This real-time water accounting allows municipal bodies to recover lost revenue, detect illegal connections immediately, and optimize their capital and operating budgets.
4	Off-Peak Domestic Waste (Overhead Tank Overflows & Open Night-Time Taps)	Rapidly growing urban centers face acute water stress, which is aggravated by preventable off-peak domestic water wastage. In cities like Indore,	The system implements an automated night-time continuous flow oversight mechanism. During off-peak night hours (when domestic water consumption should

S.No	India Level Problem	National Data / Proof	How Our Project Solves It
		thousands of liters of drinking water are wasted daily during non-peak hours when overhead residential tanks overflow or residential taps are left open overnight.	normally be zero), if any household YF-S201 sensor registers continuous water flow for an unusually prolonged period, the ESP32 identifies this as an open tap or overflow. It triggers a dual response: a localized warning buzzer sounds to alert the household, and a direct Firebase app alert is sent to both the resident's mobile app and municipal authorities to ensure rapid corrective action.
5	Delayed Response & High Dependence on Manual Labor	Traditional leak detection is slow and reactive, depending on physical ground subsidence sightings or manual valve operations. Due to severe municipal manpower shortages, finding an underground pipeline leak can take days or weeks, resulting in enormous water wastage.	The system establishes completely autonomous, tech-driven telemetry. The ESP32 is the "brain," the sensors are the "eyes/sensory nerves," the solenoid valve is the "hands," and the Firebase cloud is the "messenger." Rather than relying on manual field inspections, the system automatically detects the leak, shuts off the flow via the solenoid valve in 2-3 seconds, sounds a buzzer, and sends instant alerts with exact zone data. This

S.No	India Level Problem	National Data / Proof	How Our Project Solves It
			completely eliminates manual inspection lag and enables maintenance crews to carry out targeted repairs immediately.

6. Future Scalability and Advanced Features

While our current small-scale prototype uses standard Wi-Fi and Firebase for sending alerts, scaling this system up for real-world municipal infrastructure requires adapting to outdoor environments where street-level Wi-Fi is unavailable. For large-scale public installations, the communication network would transition to long-range wireless technologies such as LoRaWAN (Long Range Wide Area Network) or GSM SIM modules. These technologies allow the sensor nodes to transmit data over several kilometers directly to the central servers, ensuring uninterrupted remote communication even in areas without local internet access. Additionally, locating an underground pipeline leak manually across a vast neighborhood can still be challenging and slow. To make the system fully automated and efficient, a GPS module can be integrated into each pipeline sensor node. When a mismatch is detected, the system will instantly send the exact geographic coordinates of the affected segment directly to the local authority's mobile app. This eliminates the need for manual search operations, allowing municipal repair teams to target the precise leak location immediately, reducing both water waste and labor time.

7. Estimated Prototype Budget & Cost Efficiency

To keep initial proof-of-concept costs minimal, the prototype relies entirely on accessible, off-the-shelf electronics. This highlights how easily municipalities can roll out real-time monitoring without requiring heavy upfront infrastructure investment.

Itemized Prototype Cost Breakdown

- **ESP32 Microcontroller Board:** ₹450
- **Water Flow Sensors (YF-S201 x 2):** ₹600 (₹300 each)

- **Motorized Solenoid Valve (12V): ₹450**
- **Relay Module (5V/12V Channel): ₹120**
- **Power Adapter & Regulation Circuitry: ₹300**
- **Protective Enclosure (IP67 box & connections): ₹350**
- **Buzzer & Miscellaneous Jumper Wires: ₹150**
- **Total Estimated Cost: ₹2,420**

This highly economical setup acts as a functional proof-of-concept. On a mass scale, production costs are anticipated to drop by an additional 30–40%, presenting a massive cost benefit over commercial industrial-grade monitoring equipment.

8. Conclusion & Expected Societal Impact

The developed IoT framework demonstrates a highly effective, automated approach to modernizing municipal water distribution infrastructure. By delivering real-time volumetric analysis and rapid alerts, the solution transitions water management from reactive troubleshooting to continuous, data-driven protection.

Alignment with National Infrastructure Goals

- **AMRUT 2.0 (Atal Mission for Rejuvenation and Urban Transformation):** This project aligns directly with India's national mission to make urban cities water-secure. By dramatically reducing Non-Revenue Water (NRW) physical losses, it aids municipalities in maximizing tap-water efficiency and infrastructure sustainability.
- **Viksit Bharat 2047:** Preserving water and employing smart, automated public utility management directly drives India's vision of becoming a technologically advanced, resilient, and developed nation by 2047.

Expected Societal and Environmental Impact

- **Preservation of Vital Freshwater:** Mitigating invisible underground leaks preserves vital freshwater reserves, preventing artificial supply shortages in vulnerable areas.
- **Infrastructure Safety:** Prevents road cave-ins and structural damage caused by continuous, unmonitored underground leakages.
- **Equitable Distribution:** Curbing unauthorized taps ensures fair and equitable water distribution across communities.

- **Preventing Domestic Wastage:** Automated night-time flow monitoring eliminates silent, continuous wastage from unmonitored tanks and domestic fixtures.

Overall, this low-cost, scalable system empowers local authorities to conserve precious natural resources, optimize municipal operating budgets, and enhance infrastructure resilience for growing communities.

9. Project Team & Acknowledgments

This project was successfully designed, developed, and prototyped through the collective dedication and collaborative efforts of the student project team.

Student Project Team

- **Ayush Kumar Rathore**
- **Ashi Sharma**
- **Kaushikee Yadav**
- **Samarjeet Singh**
- **Harshit Kushwaha**
- **Anurag Rajput**

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